

Peter Yeshua J. Sotomango

+97455382177 | sotomango.23@outlook.com | linkedin.com/in/pysotomango23 | www.petersotomango.com

EXPERIENCE

AI and Software Engineer

Jan. 2026 – Present

Sports Vector LLC, Qatar Science and Technology Park

Doha, QA

- Implemented and maintained comprehensive observability and error monitoring across platforms, reducing errors by **80%** and achieving improved system reliability, faster incident detection, and reduced user-facing bugs.
- Architected backend system design improvements that enhanced service scalability and maintainability, enabling the platform to handle increased load while reducing deployment complexity and technical debt.
- Ensured cross-platform operability between web (*React*) and mobile (*React Native*) by aligning system architecture and UX patterns, achieving feature parity and seamless user experience across all platforms.
- Designed and administered internal tools and infrastructure that reduced manual administrative overhead by **40%**, boosting developer productivity and enabling the team to focus on core feature development.

Full Stack Developer

Dec. 2025 – Apr. 2026

CCIT Help Center, University of Doha for Science and Technology

Doha, QA

- Built a scalable **multi-tenant room booking platform** using *TypeScript, Bun, Next.js, and Supabase* that serves **500+ students** with separate frontends for students and admins connected via a centralized API, streamlining help center operations.
- Implemented **real-time synchronization** capabilities using *Supabase real-time functionality*, enabling instant booking updates across student and admin interfaces and **reducing scheduling conflicts by 40%**.
- Architected a centralized API server managing communication between student and admin interfaces, ensuring seamless data flow and real-time updates that improved system maintainability and scalability.

Research Assistant

Nov. 2025 – Apr. 2026

CCIT Student Research Program, University of Doha for Science and Technology

Doha, QA

- Conducted comprehensive literature reviews and analysis on **privacy vulnerabilities in large language models**, synthesizing current research to provide actionable insights for privacy-preserving AI implementations.
- Developed a functional **computer vision prototype for livestock weight prediction** by building image processing models and analyzing large datasets, creating a novel non-invasive estimation approach.
- Integrated machine learning techniques, data visualization tools, and experimental methodologies across **AI privacy and computer vision research domains**, producing well-rounded contributions to multiple research areas.

Student Assistant

Sep. 2023 – Apr. 2026

CCIT Help Center, University of Doha for Science and Technology

Doha, QA

- Supported students directly during lab sessions, providing **1:1 guidance** on course concepts, assignments, and offering peer tutoring sessions (**500+ students served**).
- Facilitated the CCIT Help Center by managing bookings, supporting walk-ins/appointments, and ensuring compliance with the center's policies.
- Led **recap/review sessions for 100+ students**, reinforcing key topics to strengthen understanding and improving overall academic performance.

Research Assistant

July 2025 – Oct. 2025

Columbia University & University of Doha for Science and Technology

Doha, QA

- Conducted desk research on the **implementation of AI in Qatar's healthcare sector**, analyzing adoption trends, infrastructure readiness, and regulatory context.
- Evaluated the region's **digital infrastructure**, including health data exchange platforms, cloud providers, and training centers, to assess AI integration potential.
- Collaborated with an international research team to produce insights on how **emerging technologies can improve healthcare delivery and policy in Qatar**.

Full Stack Developer

Feb. 2025 – June 2025

Cleano L.L.C

Doha, QA

- Built an **internal admin-management system** in *Next.js and Firebase* that handles partner integrations, version control, maintenance-mode toggles, and role-based admin access.

- Established **GitHub Actions CI/CD pipelines** to automatically build, lint, and test on every push to production environments, **reducing deployment errors by around 80%** and accelerating releases.
- **Optimized and rewrote the API authentication flow** by implementing **token caching and simplifying validation logic**, reducing average API latency by **50%**.

Student Volunteer

2022 – 2024

University of Doha for Science and Technology

Doha, QA

- **Supported 10+ large-scale university events** (Open Day, Orientation, Graduation, Career Fair, etc.), assisting **over 1000+ attendees** with registration, campus tours, logistics, and guest services.
- **Led and coordinated 15+ volunteers** at an international AI symposium, delegating tasks and ensuring smooth event flow across multiple venues.
- Assisted university leaders, prospective students, and parents with inquiries while also **curating official social media content** to promote engagement.

Hardware Lab Assistant

March 2023 – Sep. 2023

University of Doha for Science and Technology

Doha, QA

- **Upgraded hardware components on 100+ CCIT lab systems**, installing RAM, processors, and peripherals to maintain optimal performance.
- **Maintained software on 100+ CCIT workstations** by applying OS updates and configuring software configurations to ensure system reliability.

EDUCATION

University of Doha for Science and Technology

Doha, QA

B.Sc in Data Science and Artificial Intelligence

July 2022 – July 2026

- CGPA: 3.43/4
- Relevant Coursework: Machine Learning, Parallel and Distributed Computing, IoT Application Development, Data Visualization and Analytics, Deep Learning, Applied Data Mining

PROJECTS

ACM Learning Portal - ACM @ UDST Hackathon 1st Place Winner | *Web Development* Oct. 2025 – Present

- Led my team in conceptualizing and developing a gamified organization hub for the new UDST ACM Student Chapter.
- Designed the web application that powers the entire learning hub focusing on the gamified aspects of the platform through the use of points, achievements, badges, and more.

Intramap - University Capstone Project | *AI Development* Aug. 2025 – Present

- Designed an Indoor Localization system to help support navigation through complex indoor environments.
- Led and organized a team of 6 members to ensure tasks are easily completed and coordinated with the group supervisor with additional queries and scheduling.

TupperCare - Qatar Innovation Program Finalist | *App Development & IOT Development* March 2025 – Aug. 2025

- Led a multidisciplinary team in developing TupperCare, a smart food container that integrates sensors, microcontrollers, and a mobile app to monitor freshness and predict spoilage using machine learning.
- Designed and prototyped the hardware architecture and mobile application, selecting sensors, and building the initial app/device prototype.

WristAssured - Lifelines 2025 @ CMUQ 5th Place Winner | *IOT Development* Feb. 2025

- Built an IoT wristband prototype using Arduino/ESP32 and LoRa mesh networking to capture and transmit GPS coordinates of disaster victims to an offline-ready Raspberry Pi hub.
- Co-developed a React Native (Expo) mobile app that visualizes victim locations, assigns rescue squads, and prevents duplicate rescues to improve efficiency during crisis response.

GreenHealth AI – Microsoft & National Planning Council Datathon | *AI Development* Feb. 2025

- Conducted data analysis on the National Planning Council's Air Quality dataset to assess correlations between gas flare emissions and air quality levels.
- Utilized machine learning and data visualization techniques to identify key pollution patterns and potential mitigation strategies.

FloodAlert - ACT28 AI for Climate Hackathon | *AI Development* May 2024

- Designed and prototyped Flood Alert, an AI-powered service leveraging neural networks, satellite image recognition, and real-time data to detect flooded zones and integrate alerts into navigation apps (e.g., Google Maps, Waze).
- Demonstrated potential to reduce e-waste from EV battery failures during floods, addressing both climate resilience and sustainability challenges.

CHARACTER REFERENCES

Dr. Bekir Ciftler

Data Science and Artificial Intelligence Department Head, UDST

- Email: bekir.ciftler@udst.edu.qa
- Phone Number: 44952403

Mohamed Salah Amine Benouar

Data Science and Artificial Intelligence Lecturer, UDST

- Email: mohamedsalahamine.benouar@udst.edu.qa
- Phone Number: 44952162

TECHNICAL SKILLS

Languages: Python, Javascript, SQL, Typescript, Java, HTML/CSS

Frameworks: React, React Native, Node.js, Express

Developer Tools: Git, Cursor, Claude Code, Visual Studio Code, Warp, Postman

Cloud Platforms: Google Cloud Platform, Microsoft Azure, Oracle Cloud, Firebase, Supabase

Libraries: pandas, NumPy, Matplotlib, Seaborn, Pytorch, Tensorflow, HuggingFace

Artificial Intelligence: Machine Learning, Data Visualization, Deep Learning, Data Mining

Design Tools: Figma, CapCut, Affinity Studio, Canva

Internet of Things: Arduino, Micropython, ESP32, Sketch